Project Title

Customer Purchase Prediction Using Logistic Regression and Random Forest

Project Report submitted for the partial fulfilment of grade for the course

Machine Learning (25SC2107E)

in

A.Y. 2026-2027

Submitted by

P. ANUDEEP(2520030087)

G. ROHITH(2520030220)

Team Number:

Section: 9

Under the guidance of

Ms. Rani Medidha , Associate Professor, Department of Computer Science and Engineering

Hyderabad-500090, Telangana, India.

Problem Statement:

In e-commerce platforms, a large number of customers visit and browse products, but only some of them complete a purchase. It is difficult for businesses to manually identify customers who are more likely to buy a product because of the large amount of customer and behavioral data available.

The main problem of this project is to develop a **Machine Learning-based customer purchase prediction system** that can classify customers as likely to purchase or not likely to purchase. **Logistic Regression and Random Forest** are used to build prediction models based on customer information and online activities. The performance of these algorithms is compared to identify the more effective model for customer purchase prediction.

Abstract:

Customer purchase prediction is an important application of Machine Learning in the e-commerce sector. E-commerce platforms generate large amounts of customer data through product views, browsing activities, previous purchases, cart activity, and other interactions. This project aims to predict whether a customer is likely to make a purchase using **Logistic Regression and Random Forest** algorithms. The customer data is preprocessed and used to train both models, and their performance is compared using evaluation measures such as accuracy, precision, recall, and F1-score. The prediction system can help e-commerce businesses identify potential customers, improve marketing strategies, provide personalized recommendations, and increase sales.

Introduction

The growth of e-commerce has changed the way people search for and purchase products. Online shopping platforms collect various types of customer information, including browsing history, products viewed, cart activity, previous purchases, and interactions with offers. Analyzing this information can help businesses understand customer interests and purchasing behavior.

Machine Learning provides effective techniques for identifying patterns in customer data and making predictions. In this project, customer purchase prediction is considered a **binary classification problem**, where the output indicates whether a customer will make a purchase or not.

Two classification algorithms, **Logistic Regression and Random Forest**, are used in this project. Logistic Regression is simple and effective for binary classification, while Random Forest uses multiple decision trees to handle complex patterns in customer data. By comparing the performance of both models, the project aims to select a suitable approach for predicting customer purchases. The resulting system can support targeted marketing, personalized recommendations, better customer engagement, and improved e-commerce sales.

Signature of the Guide